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# ADMESH: An advanced, automatic unstructured mesh generator for shallow water models

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**Abstract** In this paper, we present the development and application of a two-dimensional, automatic unstructured mesh generator for shallow water models called ADMESH. Starting with only target minimum and maximum element sizes and points defining the boundary and bathymetry/ topography of the domain, the goal of the mesh generator is to automatically produce a high-quality mesh from this minimal set of input. From the geometry provided, properties such as local features, curvature of the boundary, bathymetric/topographic gradients, and approximate flow characteristics can be extracted, which are then used to determine local element sizes. The result is a high-quality mesh, with the correct amount of refinement where it is needed to resolve all the geometry and flow characteristics of the domain. Techniques incorporated include the use of the so-called signed distance function, which is used to determine critical geometric properties, the approximation of piecewise linear coastline data by smooth cubic splines, a so-called mesh function used to determine element sizes and control the size ratio of neighboring elements, and a spring-based force equilibrium approach used to improve the element quality of an initial mesh obtained from a simple Delaunay triangulation. Several meshes of shallow water domains created by the new mesh generator are presented.

**Keywords** Mesh generation · Shallow water modeling · DistMesh

## 1 Introduction

Computational domains used in shallow water modeling are generally large in extent and possess complex geometry introduced by intricate coastlines and steep bathymetric gradients. A large range of scales exist within these shallow water domains, ranging from kilometers in the deep ocean to tens of meters in the near shore. As a result of these varying scales and the complex geometry, it is advantageous to use unstructured meshes to discretize the domain. Specifically, smaller mesh elements are required in the domain to resolve the curvature of the coast, channels, and tributaries, as well as any complex flow characteristics, while larger mesh elements should be used wherever possible to maximize computational efficiency. Ideally, the generation of such a mesh should be automated, requiring minimal user input and interaction.

There has been extensive work in the general field of automatic mesh generation. Here, with one notable exception, we limit the discussion to mesh generation work in the context of shallow water modeling. Hagen et al. (2000, 2001, 2002), generate triangular finite element meshes through utilization of a local truncation error analysis (LTEA), in which mesh element sizes are

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controlled by a posteriori error estimates. A simulation on a coarse mesh is performed, and then a posteriori error calculations are made and mesh elements are generated which keep the estimated local truncation errors below some prescribed tolerance. Bilgili et al. (2006) generate unstructured triangular meshes via a mesh generator called BatTri, which consists of a graphical mesh editing interface coupled with several bathymetry-based refinement algorithms designed to check and improve mesh quality. Gorman et al. (2006, 2008) use an error metric based on bathymetry data along with a polyline approximation of the boundary to generate unstructured anisotropic triangular meshes. The error metric is based on the use of the Hessian, and mesh elements are generated that keep the error metric below some prescribed tolerance. Lambrechts et al. (2008) also utilize the Hessian along with cubic B-splines to generate meshes in spherical coordinates via a state-of-the-art CAD program called Gmsh. Another popular meshing tool among shallow water modelers is the Surface Water Modeling System (SMS) developed by Zundel and Jones (1996), which has an extensive set of graphical mesh generation and editing tools. While meshes produced using SMS can be of high quality, the mesh generation and editing process can be both user-intensive and time-consuming. Finally, although not specific to the area of shallow water modeling, we mention the automatic mesh generation work of Persson and Strang (2004), Persson (2005, 2006), and in particular his DistMesh program—a simple, open-source mesh generator implemented in Matlab (and available online Persson 2012), which provides the basis and serves as the main inspiration for the current work.

Building and expanding on some of the work discussed above, in this paper, we present the development and application of an advanced mesh generator, called Admesh, that is capable of automatically producing high-quality unstructured meshes for shallow water domains starting with only target minimum and maximum element sizes and points defining the boundary and bathymetry/topography of the domain. The automatic mesh generator incorporates some of the latest and most advanced techniques recently developed in the area of mesh generation, as well as novel approaches to incorporate factors that are specific to the problem of shallow water flow. Some of the techniques that are incorporated include a novel approach we have developed to compute exact signed distance functions, which are widely used in the context of level set methods, the approximation of piecewise linear coastline data by smooth cubic splines, which are used to compute shoreline curvature, and a so-called mesh size function, which is used to determine element sizes and control the size ratio of neighboring elements. The mesh generation approach is implemented in Matlab and uses an interactive graphical user interface (GUI) that allows the user to view and edit input data and produce relevant output. The GUI also contains an option to extract coastline data via the National Oceanic and Atmospheric Administration's (NOAA) web-based coastline extractor database.

The remainder of this paper is organized as follows. The main ideas behind the methods and algorithms employed by Admesh are briefly described, and its capabilities are demonstrated in a series of examples, where a number of meshes created by Admesh are presented and assessed in terms of element quality. Finally, we end the paper with some conclusions and future work.

## 2 Methods and algorithms

### 2.1 Domain representation

Prior to describing the meshing process, we first describe how the domain is represented within the Admesh code. To begin, consider a given shallow water domain $\Omega \subset \mathbb{R}^3$, which can be expressed as $\Omega = \Omega_{xy} \times z$, where $z(x, y)$ denotes the vertical elevation (bathymetric depth or topographic height) of a point $(x, y) \in \Omega_{xy}$. We denote the boundary of $\Omega_{xy}$ by $\partial\Omega_{xy}$, which, along with $z$, is generally known at a discrete number of points obtained from, for example, coastline and bathymetric/topographic databases; see, for example, NGDC, Coastline Extractor (n.d.); NGDC, Bathymetry & Global Relief (n.d.), respectively.

In Admesh, the boundary $\partial\Omega_{xy}$ is input via a list of *ordered* points given in either geographic (latitude, longitude) or standard Cartesian coordinates. The Admesh GUI contains options to obtain boundary data from either an existing mesh or from NOAA's online Coastline Extractor tool (NGDC, Coastline Extractor n.d.). These points are automatically joined in Admesh to form a closed polygon that represents the external boundary of the domain. Internal boundaries, e.g., islands or internal structures such as weirs (see, for example, Dawson et al. 2011), are also input via a list of ordered points and are joined to form closed polygons. The list of boundary points and resulting polygons are checked for various "inconsistencies" prior to meshing, e.g., the presence of double points, inconsistent direction among polygons, etc., which are corrected as appropriate. In addition to making use of this piecewise linear boundary representation, Admesh also utilizes a smooth (twice-differentiable) cubic spline approximation of the boundary (see Fig. 4), which allows for boundary curvature calculations to be performed as described in the next section. Similar to the boundary representation, the vertical extent of the domain $z$ is input into ADMESH via a (generally nonuniformly spaced) set of data points, which can also be obtained from either an existing mesh of the domain or a separate bathymetric/topographic data set.

Given these boundary and bathymetric/topographic representations, discretization of the domain is essentially defined through the use of two discrete functions defined over a (structured) background mesh or grid that covers the entire extent of the domain. These two discrete functions are (a) a signed distance function $\phi(\mathbf{x})$, which gives the shortest distance from a background grid point $\mathbf{x} = [x, y]$ to the boundary; and (b) a bathymetry function $B(\mathbf{x})$, which gives the corresponding bathymetric depth of the point (without loss of generality, we refer to this as a bathymetry function, though it would also describe topographic height for portions of the domain that cover land). As described in the next section, a number of related functions and quantities that play an important role in controlling the mesh sizes $h$ throughout the domain are computed from these two discrete functions.

*Remarks*

(a) The signed distance function is a concept that is widely used in the context of level set methods; see, for example, Sethian (1999). Points located on the boundary of the domain have a distance equal to zero, nodes outside of the domain have a positive distance value, and nodes inside the domain have a negative distance value (see Fig. 1). Distance functions can be computed using, for example, the so-called fast marching method of Sethian (1999), which gives approximate distance functions in $\mathcal{O}(N \log(N))$ time, where $N$ is the number of background grid points, or the closest point transform (CPT) of Mauch (2003), which gives exact distance functions (within machine precision) in $\mathcal{O}(N)$ time. In ADMESH, we have implemented a variant of the CPT algorithm. From the signed distance function, the so-called medial axis (or topological skeleton) of the domain, which is the set of interior points that have equal distances to two or more points on the boundary of the domain, can be computed. This, in turn, can be used to determine "local feature" sizes lfs($\mathbf{x}$)—features such as the width of narrow channels and islands. These concepts are illustrated along with the distance function in Fig. 1, while Fig. 2 shows the distance function for the Western North Atlantic tidal domain.

(b) Note that the boundary in ADMESH is not strictly defined by either the piecewise linear polygons or cubic spline representation. Instead, it is defined by the distance function $\varphi(\mathbf{x})$ stored on the structured background grid—the spacing of which determines the accuracy with which the coastline is represented (see Fig. 3). Defining the boundary in this way allows boundary nodes to move along the boundary during the iterative mesh generation process and does not require boundary elements to be a set size at a prescribed location as is common with other mesh generation approaches that fix boundary nodes. This allows the size and location of the boundary elements to be determined by the mesh size function alone, allowing for optimal placement of elements along the boundary. In ADMESH's current implementation, the background grid resolution can be set to one of three

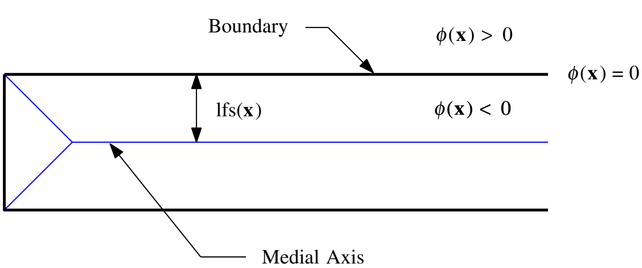


**Fig. 1** Depiction of the signed distance function $\phi(\mathbf{x})$, medial axis, and local feature size lfs($\mathbf{x}$)

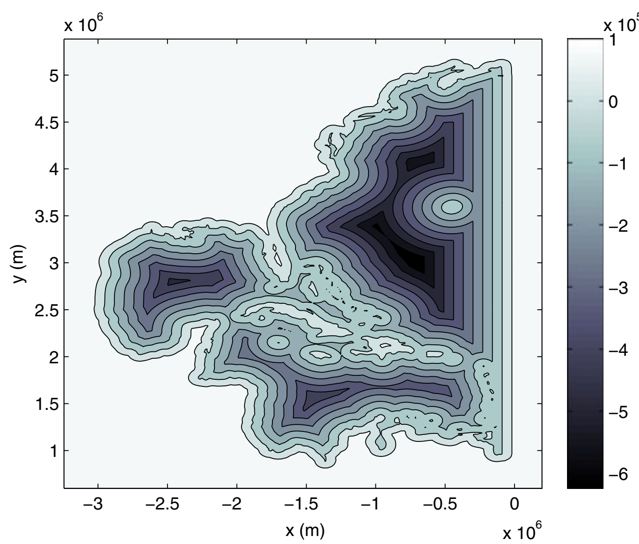


**Fig. 2** Distance function for the Western North Atlantic tidal domain

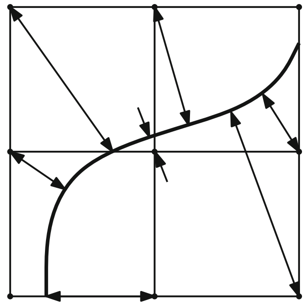

**Fig. 3** Closest distances from background nodes to boundary

settings: high, medium, or low, all of which are functions of the prescribed minimum element size.

(c) The bathymetry function $B(\mathbf{x})$ defined over the background grid is obtained in ADMESH by first fitting a triangle-based piecewise linear surface to the set of bathymetric/topographic data points input by the user and then interpolating this surface onto the points of the background grid. As described in the next section, the bathymetry function is used in considerations related to providing sufficient mesh resolution related to bathymetric gradients $\nabla B$ and tidal wave lengths $\lambda$. It should be noted that if the zero contour lines of the topography/bathymetry data are in conflict with the prescribed coastline, ADMESH gives precedence to the coastline, and the bathymetry is adjusted accordingly using a weighted average of neighboring grid points to resolve the conflict. Future versions of ADMESH will allow the user to specify which data set, i.e., the coastline or bathymetry, should take precedence in defining the boundary of the shallow water domain.

### 2.2 Mesh resolution

The use of unstructured meshes in shallow water modeling has been shown to be an efficient way to obtain highly accurate numerical solutions with minimal computational effort; see, for example, Leuttich and Westerink (1995), Westerink et al. (1994a, b). This efficiency is a direct result of the ability of unstructured meshes to easily provide varying spatial resolution throughout the domain. Providing a means by which appropriate mesh resolution or element sizes are automatically prescribed in different parts of the domain—based on various physical and geometric factors discussed below—is a critical part of the automatic mesh generation process. In many meshing algorithms, this is accomplished as an a priori procedure before the meshing process begins by constructing a mesh/element size function $h(\mathbf{x})$ defined over a background grid, from which target element sizes can be obtained via interpolation during the meshing process. Many techniques exist for generating mesh size functions; see, for example, Persson (2006), Zhu et al. (2002). ADMESH employs the basic mesh size function technique proposed by Persson (2006) with additional modifications made to incorporate factors that are specific to the problem of shallow water flow. This technique is described briefly below along with a number of factors that should be considered to determine mesh resolution, i.e., element sizes of shallow water domains. For a thorough discussion on resolution issues in numerical models of oceanic and coastal circulation see Greenberg et al. (2007). In ADMESH, mesh function inputs are based on the following criteria:

1. **Shoreline curvature**: It is important that the geometric complexity of natural shorelines be adequately represented in shallow water models; see, for example, Fig. 4. This is accomplished in ADMESH by controlling mesh resolution based on shoreline curvature. Areas of shoreline that possess a high curvature value require smaller elements to adequately resolve the geometry. Given a boundary as described in Section 2.1, its curvature $\kappa(\mathbf{x})$ can be calculated from a smooth parametric cubic spline approximation, with $\kappa(\mathbf{x})$ given by

$$\kappa = \frac{\left| S_x{}' S_y{}'' - S_y{}' S_x{}'' \right|}{\left[ S_x{}'^2 + S_y{}'^2 \right]^{3/2}} \tag{1}$$

where $S_x$ and $S_y$ are the $x$- and $y$- components of the vector spline function that passes through the set of boundary points $\{(x_i, y_i)\}_{i=1}^{n}$, where $n$ is the number of boundary points that make up the boundary polygon. Periodic boundary conditions with period $x_n - x_1$ are used to interpolate the data points due to the fact that the data points close on themselves; for further details, see Conroy (2010) or Engeln-Mullges and Uhlig (1996). It should be noted that the use of cubic splines to represent the boundary is not novel to that of ocean mesh generation, as a variant of cubic splines is utilized by Lambrechts et al. (2008) to represent coastal boundaries in Gmsh. However, utilization of the curvature of coastal boundaries as input into the mesh function is, to our knowledge, novel to that of coastal mesh generation. After calculating, the curvature ADMESH requires that

$$h_{\text{shore}}(\mathbf{x}) \le \frac{1}{K\,|\kappa(\mathbf{x})|}, \tag{2}$$

where $K$ is the number of elements per radian as specified by the user. Based on generating a

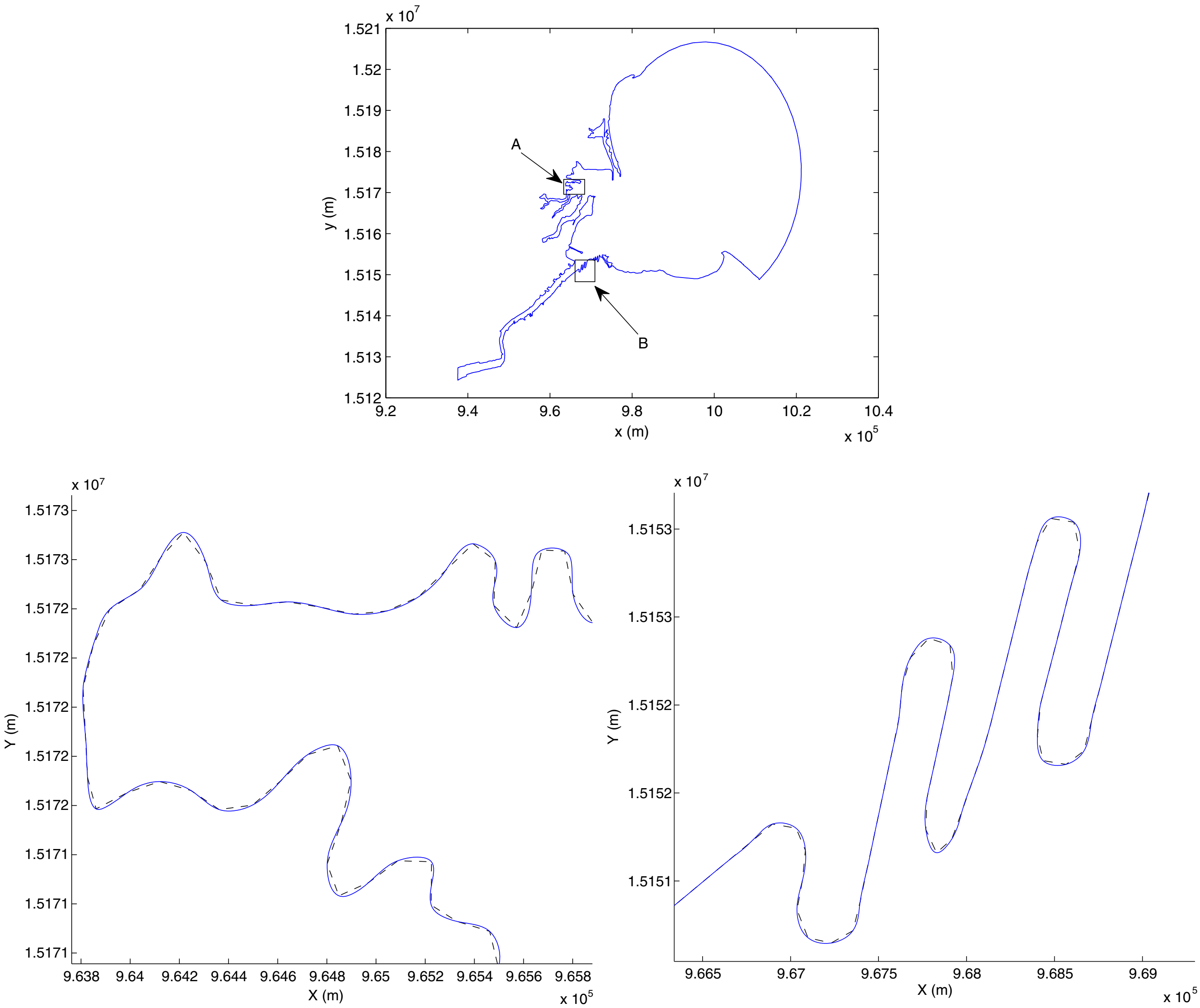


**Fig. 4** Cubic spline representation of Maumee Bay, Lake Erie. *Bottom left* is zoomed in view of *A*. *Bottom right* is zoomed in view of *B*. *Dashed line* is piecewise boundary, while *blue line* is cubic spline approximation

number of meshes using real shoreline data, we have determined that $K = 15$ to 25 seems to be a reasonable range for this parameter. It should be noted that $K = 15$ will produce a mesh with fewer elements as compared to a mesh with $K = 25$; see, for example, Fig. 5. ADMESH then defines the shoreline mesh size input function as

$$h_{\text{shore}}(\mathbf{x}) = \begin{cases} \frac{1}{K|\kappa(\mathbf{x})|} & \text{if } |\phi(\mathbf{x})| \leq 2\Delta x \\ \infty & \text{if } |\phi(\mathbf{x})| > 2\Delta x \end{cases}, \quad (3)$$

where $\Delta x$ is the resolution of the background grid. This leaves the interior nodes of the domain unaffected by shoreline curvature. Note that shoreline curvature values will be accurate for background nodes with $\phi(\mathbf{x}) = 0$ and will become less accurate as $|\phi(\mathbf{x})|$ increases. However, due to the narrow band that ADMESH restricts shoreline curvature criteria to effect, this error is insignificant. See the discussion in Persson (2005) for a method that corrects the curvature values for nodes located away from the boundary. Also, note that we set $h_{\text{shore}}(\mathbf{x}) = \infty$ for background nodes with $|\phi(\mathbf{x})| \leq 2\Delta x$, whose closest boundary polygon edge is an open-ocean boundary edge. This results in the open-ocean boundary being unaffected by curvature.

2. **Local feature size**: Adequate resolution of channels/tributaries in shallow water domains is noted in Greenberg et al. (2007) to be critical in obtaining valid solutions, especially for transport problems. ADMESH ensures this requirement is met through

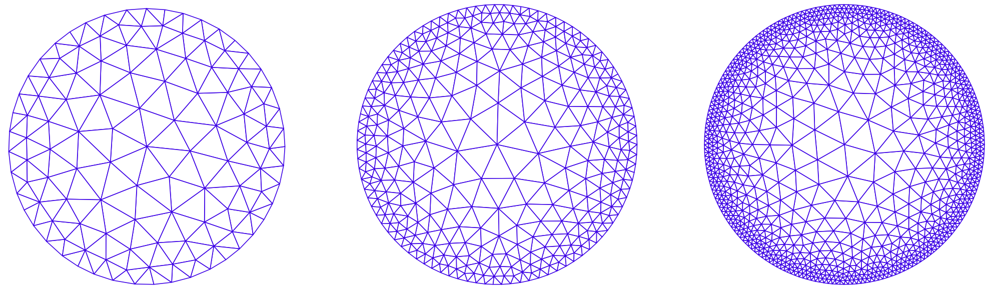

**Fig. 5** Meshes with different $K$ values. *Left* $K = 50/2\pi$, *center* $K = 100/2\pi$, and *right* $K = 200/2\pi$

the use of a local feature size mesh adaptation that controls the minimum number of elements that span the channels/tributaries. In particular, ADMESH requires that

$$h_{lfs}(\mathbf{x}) \leq \frac{\mathrm{lfs}(\mathbf{x})}{R}, \tag{4}$$

where the local feature size $\mathrm{lfs}(\mathbf{x})$ is approximately half the width of the channel/tributary at a point $\mathbf{x}$. More specifically, ADMESH defines local feature size as

$$\mathrm{lfs}\,(\mathbf{x}) = |\phi_{MA}\,(\mathbf{x})| + |\phi\,(\mathbf{x})|\,, \tag{5}$$

where $\phi_{MA}\,(\mathbf{x})$ is the distance from the the medial axis of the shallow water domain to any point $\mathbf{x}$ contained within the background grid. Defining local feature size in this fashion gives $\mathrm{lfs}(\mathbf{x}) = \phi_{MA}\,(\mathbf{x})$ at the boundary, with $\mathrm{lfs}(x)$ increasing as one gets further away from the boundary. The parameter $R$ in Eq. 4 sets the minimum number of elements that span half of the channel/tributary, e.g., if $R = 1$, then a minimum of two elements will span any channel/tributary within the domain.

*Medial axis location*

Initialization of $\phi_{MA}\,(\mathbf{x})$ requires location of the medial axis of the shallow water domain. There are numerous methods for accomplishing this, including the methods in Armstrong et al. (1991), Meshkat and Sakkas (1987), Persson (2005), or Yao and Rokne (1991). ADMESH identifies the location of the medial axis through the use of a divergence-based formulation developed by Siddiqi et al. (2002). Siddiqi et al. (2002) consider the nonlinear Eikonal equation, an equation that arises from geometric optics,

$$|\nabla\phi\,(\mathbf{x})| = F\,(\mathbf{x}) \quad \text{in} \quad \Omega, \qquad \phi = 0 \quad \text{on} \quad \Gamma \tag{6}$$

where $\Omega$ is the domain contained within $\mathbb{R}^2$ or $\mathbb{R}^3$, $F(\mathbf{x}) > 0$ is the so-called speed function, and $\Gamma$ is a prescribed curve or surface, which, in our case, is the medial axis of $\Omega$. The shocks and singularities that form in the solution to Eq. 6 serve as a means to locate medial axis points and are detected through the use of a Hamiltonian formulation of the Eikonal equation. Specifically, medial points can be distinguished from non-medial points by computing the average outward flux of the vector field $\nabla\phi\,(\mathbf{x})$ about a point $\mathbf{x}$. The average outward flux, AOF, is defined in Siddiqi et al. (2002) as

$$\mathrm{AOF} = \frac{\int_{\delta R} \langle \nabla\phi\,(\mathbf{x})\,, \mathbf{N}\rangle\, ds}{\mathrm{length}\,(\delta R)} \tag{7}$$

and can be related to the divergence via

$$\nabla \cdot (\nabla\phi\,(\mathbf{x})) \equiv \lim_{\Delta a \to 0} \frac{\int_{\delta R} \langle \nabla\phi\,(\mathbf{x})\,, \mathbf{N}\rangle\, ds}{\Delta a}, \tag{8}$$

where $ds$ is an element of the boundary $\delta R$, $\mathbf{N}$ is the outward normal of each contour point, $\Delta a$ is the area , and $\langle .,.\rangle$ is an inner product. See Siddiqi et al. (2002) for a discussion on why calculation of Eq. 8 is an effective method for calculating singularities in Eq. 6.

ADMESH detects the location of medial axis points in the background grid by first determining the average outward flux of the vector field $\nabla\phi\,(\mathbf{x})$ at each background node $\mathbf{x}_k$, i.e.,

$$\mathrm{AOF}(\mathbf{x}_k) = \frac{\sum_{i=1}^{n} \mathbf{N}_{ik} \cdot \nabla\phi\,(\mathbf{x}_{ik})}{n}, \tag{9}$$

where $\phi$ is the distance function, $\mathbf{x}_{ik}$ is a neighbor of the background node $\mathbf{x}_k$, $\mathbf{N}_{ik}$ is the outward normal of $\mathbf{x}_{ik}$, and $n$ is the number of neighbors ($n = 8$ in two dimensions). A threshold value is then set on the average outward flux, and nodes in the background grid are classified as medial or non-medial points based on their average outward flux value. It should be noted that the use of a threshold value by itself does not guarantee a thin connected set of points that define the medial axis (Persson 2005). For instance, a high threshold (a value further from zero) will result in a thin skeleton that contains gaps, while a low threshold (a value closer to zero) tends to lead to a thick skeleton with unwanted branches. However, because of the nature of the use of the medial axis in ADMESH, it is unnecessary to obtain a thin skeleton topology. Therefore, ADMESH uses a threshold value of −0.165 to identify medial axis points in the background grid regardless of domain shape or size. This value produces a skeleton that is one to three nodes thick and is devoid of holes or unwanted branches and was determined via numerous computational experiments; see, for example, Fig. 6. For a method that "thins" the skeleton produced via average outward flux calculations, the reader is referred to Siddiqi et al. (2002).

*Distance to medial axis*

The problem of calculating distances from the medial axis to the domain boundary involves solution of the Eikonal Eq. 6, with a speed function $F(\mathbf{x}) = 1$. A number of methods exist for generating solutions to Eq. 6, including level set methods (Sethian 1999), upwind viscosity methods (Rouy and Tourin 1992), and the aforementioned fast marching method (Sethian 1999). ADMESH has two options for calculating $\phi_{MA}(\mathbf{x})$: the first uses the fast marching method to generate approximate solutions to Eq. 6, while the second uses a variation of

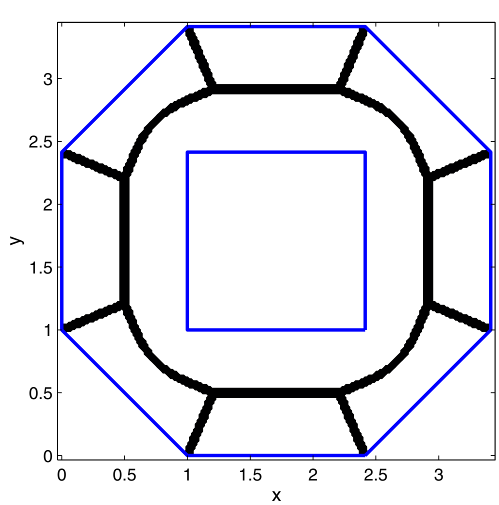


**Fig. 6** Medial axis (*black line*) of an octagon with a square hole (*blue line*). The *hole* is a representation of an island

an upwind viscosity method that builds steady state solutions to the unsteady form of Eq. 6, i.e.,

$$\frac{\partial \phi}{\partial t} + |\nabla \phi(\mathbf{x})| = F(\mathbf{x}) \quad \text{in} \quad \Omega \tag{10}$$
$$\phi = g(\mathbf{x}) \quad \text{on} \quad \partial\Omega$$
$$\phi = 0 \quad \text{on} \quad \Gamma$$

where, in this context, $\partial\Omega$ is the background grid boundary and $\Gamma$ is the medial axis. ADMESH obtains a steady state solution to Eq. 10 via iteration using explicit Runge–Kutta methods in time and an entropy satisfying upwind scheme in space. Usually such a scheme would require $\mathcal{O}(N^3)$ operations; however, the number of iterations to convergence is greatly reduced by the handling of the boundaries and initial condition on the background grid. Specifically, we calculate the exact distance from the medial axis to the boundary of the background grid $\partial\Omega$. Then, the initial condition of the background nodes are defined as

$$\phi_{MA_{IC}}(\mathbf{x}) = r_k + \phi(\mathbf{x}) \tag{11}$$

where $r_k$ is the absolute value of the distance function of a medial point who shares the same boundary polygon edge $k$ as the given background node. The use of Eq. 11 coupled with the treatment of the boundary reduces the cost of the iterative method to approximately $\mathcal{O}(N^2)$. It should be noted that the iterative procedure produces a $\phi_{MA}(\mathbf{x})$ that is more accurate than the fast marching method; therefore, the iterative procedure is used to create high resolution meshes in the near shore, while the fast marching method is used to create tidal meshes. After the medial distance is determined, the local feature size input function is then defined as

$$h_{lfs}(\mathbf{x}) = \begin{cases} \frac{lfs(\mathbf{x})}{R} & \text{if } \phi(\mathbf{x}) \le 0 \\ \infty & \text{if } \phi(\mathbf{x}) > 0 \end{cases}. \tag{12}$$

Note that, to our knowledge, the use of automated local feature size calculations is novel to that of coastal mesh generation.

3. **Bathymetry gradients**: To ensure that adequate resolution is used in the domain to capture changing bathymetry, as suggested in Legrand et al. (2007), we require that

$$h_{\text{bathy}}(\mathbf{x}) \;\le\; s\left|\frac{\nabla B(\mathbf{x})}{B(\mathbf{x})}\right|^{-1}, \tag{13}$$

where $s$ is a user-specified parameter (taken to be in the range $s = 0.05$–$0.30$ in Legrand et al. 2007). Use of the length scale $|\nabla B(\mathbf{x})/B(\mathbf{x})|^{-1}$ is motivated by the fact that it appears in many analytical studies on shelf break dynamics; see, for example, Huthnance (1995), Loder (1980), and Xing and Davies (1998). ADMESH then defines the bathymetry mesh size over the background grid as

$$h_{\text{bathy}}(\mathbf{x}) = \begin{cases} s\left|\frac{\nabla B(x)}{B(x)}\right|^{-1} & \text{if } \phi(\mathbf{x}) \le 0 \\ \infty & \text{if } \phi(\mathbf{x}) > 0 \end{cases}. \tag{14}$$

4. **Tidal wavelengths**: Assuming that tidal waves propagate at their linear shallow water wave speeds $c$, i.e., $c = \lambda/T = \sqrt{gB}$, where $\lambda$ is the tidal wavelength, $T$ is the wave period, and $g$ is the acceleration of gravity, ADMESH requires that

$$h_{\text{tide}}(\mathbf{x}) \;\le\; \frac{\lambda}{C}, \tag{15}$$

to guarantee that proper mesh resolution is used to capture the flow characteristics of the shallow water domain. In Eq. 15, $C$ is the number of elements per wavelength as specified by the user. A detailed study of modeling tides in the western North Atlantic using unstructured meshes suggests a value of $C = 25$ for developing meshes (Westerink et al. 1994b). ADMESH defines the tidal mesh size input over the background grid as

$$h_{\text{tide}}(\mathbf{x}) = \begin{cases} \frac{\sqrt{gB(\mathbf{x})}T}{C} & \text{if } \phi(\mathbf{x}) \le 0 \\ \infty & \text{if } \phi(\mathbf{x}) > 0 \end{cases}. \tag{16}$$

5. **Minimum and maximum resolution**: Finally, the user can explicitly specify minimum and maximum element sizes $h_{\min}$ and $h_{\max}$, respectively, between which all element sizes of the initial mesh $h_0(\mathbf{x})$ must fall, i.e.,

$$h_{\min} \;\le\; h_0(\mathbf{x}) \;\le\; h_{\max}. \tag{17}$$

We note that any of the above arguments can be turned on or off when running ADMESH.

The mesh size function $h_0(\mathbf{x})$ designed to meet the set of requirements outlined above (or some subset of them) is obtained by setting

$$h_{IC}(\mathbf{x}) = \min\left(h_{\text{shore}}(\mathbf{x}), h_{\text{lfs}}(\mathbf{x}), h_{\text{bathy}}(\mathbf{x}), h_{\text{tide}}(\mathbf{x})\right) \tag{18}$$

and then

$$h_0(\mathbf{x}) = \max\left(h_{\min}, h_{\text{IC}}(\mathbf{x})\right). \tag{19}$$

Note that this condition restricts the minimum element size to that prescribed by the user. The resulting mesh size function, $h_0(\mathbf{x})$, is then the initial condition for a partial differential equation that we solve to obtain a final mesh size function $h(\mathbf{x})$ designed to meet an additional "grading" requirement. The use of properly graded meshes has been shown to be an important consideration in shallow water modeling; see, for example, Westerink et al. (1994a). Specifically, we require that the ratio of neighboring element sizes is less than a factor $G$ (specified by the user). In a continuous setting, this can be expressed as

$$|\nabla h(\mathbf{x})| \;\le\; g \approx G - 1. \tag{20}$$

This condition can be met by seeking a steady state solution to the so-called gradient limiting equation,

$$\frac{\partial h}{\partial t} + |\nabla h| = \min\left(|\nabla h|, g\right) \tag{21}$$

with an initial condition

$$h(\mathbf{x}, t = 0) \;=\; h_0(\mathbf{x}). \tag{22}$$

Note that at steady state ($\partial h/\partial t = 0$), Eq. 21 reduces to $|\nabla h| = \min(|\nabla h|, g)$, which is equivalent to the condition in Eq. 20; see Persson (2005) for more details. ADMESH obtains a steady state solution to Eq. 21 via the iteration

$$h_{ij}{}^{n+1} = h_{ij}{}^{n} + \Delta t\left(\min\left(\nabla_{ij}{}^{+}, g\right) - \nabla_{ij}{}^{+}\right) \tag{23}$$

with

$$\nabla_{ij}{}^{+} = [\max\left(D^{-x}h_{ij}{}^{n}, 0\right)^2 + \min\left(D^{+x}h_{ij}{}^{n}, 0\right)^2 + \max\left(D^{-y}h_{ij}{}^{n}, 0\right)^2 + \min\left(D^{+y}h_{ij}{}^{n}, 0\right)^2]^{1/2},$$

where $D^{-x}$ is the backward difference operator in the $x$-direction, $D^{+x}$ is the forward difference operator in the $x$-direction, etc. Figure 7 shows the solution to the gradient limiting Eq. 21 with initial condition Eq. 22 for the Western North Atlantic.

### 2.3 Mesh generation

Given the domain representation and the mesh size function as described above, a mesh is generated for a given domain by first performing a Delaunay triangulation of an initial set of mesh nodes that has a density proportional to $1/h(\mathbf{x})^2$; i.e., more nodes are placed in areas requiring high mesh resolution (small $h$) as prescribed by the mesh size function. The location of the nodes of this initial mesh is then refined iteratively by making use of some simple concepts from spring mechanics as described briefly below.

The notion of incorporating ideas from spring mechanics to aid in the process of mesh generation dates back to (at least) the work of Gnoffo (1982, 1983) and has since been used in a number of mesh generation and adaption algorithms; see, for example, Bossen and Heckbert (1996), Bourguignon and Cani (2000), Palmerio (1994), Persson and Strang (2004). The basic idea is

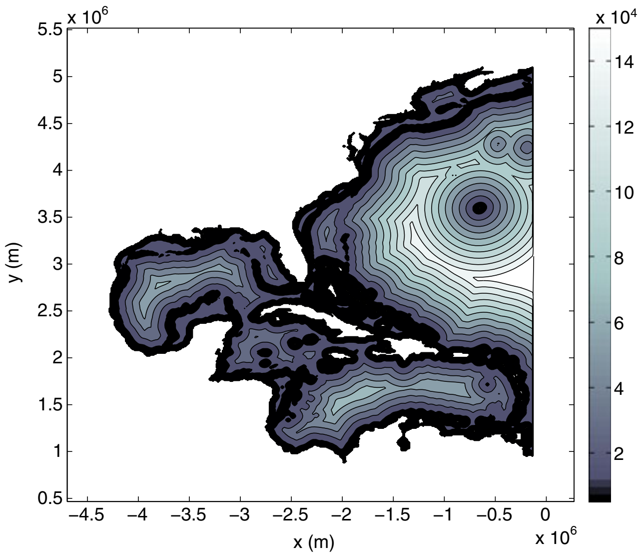


**Fig. 7** Mesh size function $h(\mathbf{x})$ for the Western North Atlantic tidal domain

to model each element edge of the mesh as a spring with, for example, a force of the form $f = k(l_h - l_i)$, where $k$ is a spring constant, $l_i$ is the length of the element edge in the current triangulation $T_i$, and $l_h$ is the desired length as prescribed by the mesh size function $h(\mathbf{x})$. A sum of spring forces is then taken around each mesh node, and force equilibrium is sought by allowing mesh nodes to be repositioned *and* by allowing the mesh topology to change through re-triangulation until the entire spring system (i.e., the entire mesh) is in equilibrium (to within a prescribed tolerance). This is accomplished in ADMESH by searching for a steady state solution to the system of ordinary differential equations given by

$$\frac{\mathrm{d}\mathbf{p}}{\mathrm{d}t} = \mathbf{F}\left(\mathbf{p}\right), \tag{24}$$

where $\mathbf{p}$ is the vector of the unstructured mesh node coordinates $(x,\ y)$ and $\mathbf{F}\left(\mathbf{p}\right)$ is the vector of the sum of the forces $f = k(l_h - l_i)$ at each node. If a steady state solution is found, then $\mathbf{F}\left(\mathbf{p}\right) = 0$, and the system is in equilibrium. ADMESH obtains a steady state solution to Eq. 24 via the forward Euler method,

$$\mathbf{p}_{n+1} = \mathbf{p}_n + \Delta t \mathbf{F}\left(\mathbf{p}_n\right), \tag{25}$$

where $\Delta t$ is based on a CFL condition. Following the methodology of Persson and Strang (2004), if a mesh node moves outside of the domain (as determined by the sign of the distance function) while solving for equilibrium, it is "pushed" back to the closest point on the boundary using the gradient of the distance function. Figure 8 shows a mesh generated using this approach, where both the initial Delaunay triangulation of the domain and the final mesh obtained from force equilibrium are shown. As demonstrated in the

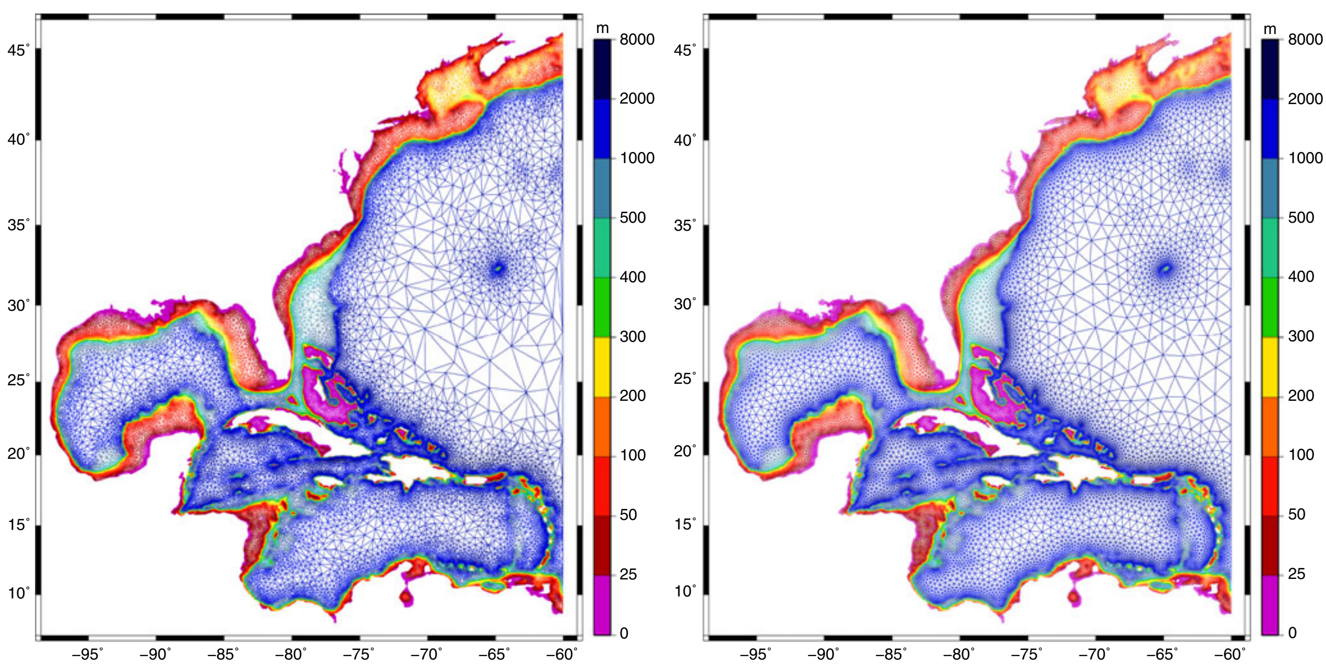


**Fig. 8** A mesh of the Western North Atlantic, Gulf of Mexico, and Caribbean Sea created using ADMESH showing the initial Delaunay triangulation of the domain (*left*) and the final mesh obtained from force equilibrium (*right*). *Colors* indicate bathymetric depths

next section, this approach produces very high-quality meshes with respect to certain mesh quality measures.

## 3 Example meshes

In this section, meshes of several shallow water domains created using ADMESH are presented. For each domain, coastline data was obtained either from NOAA's online Coastline Extractor tool (NGDC, Coastline Extractor n.d.) or an existing mesh. Each mesh was generated on a 64-bit DELL Optiplex 990 machine with 12-GB RAM and dual quad-core 3.40-GHz Intel i7-2600 processors at the Computational Hydrodynamics and Informatics Laboratory at The Ohio State University. For each mesh, we report the time required to create the mesh (CPU time) and measures of the mesh quality, which are described below.

### 3.1 Mesh quality

There are many measures that can be used to assess the quality of a mesh; see, for example, the survey paper by Field (2000). One commonly used measure, and the one used here, is twice the ratio of the radius of the largest circle contained in the triangular element $r_\mathrm{i}$ (the so-called incircle or inscribed circle of the triangle) and the radius of circumscribed circle of the triangle $r_\mathrm{o}$, i.e.,

$$q = 2\left(\frac{r_\mathrm{i}}{r_\mathrm{o}}\right) = \frac{(b+c-a)\,(c+a-b)\,(a+b-c)}{abc}$$

where $a$, $b$, and $c$ are the edge lengths of the triangular element. Note that this measure gives a value of $q = 1$ for an equilateral triangle and $q = 0$ for a completely degenerate triangle. This measure is illustrated graphically in Fig. 9.

It is desirable to have a mesh composed of nearly equilateral triangles when using finite element methods because upper bounds on errors generally depend on the smallest angle in the mesh, and good numerical results are usually obtained for elements whose angles are close to 60° (Field 2000). Here, we report the mean and minimum values of the mesh quality measure $q$ for a given mesh. Typically, meshes created with ADMESH have a mean element quality measure of $q \geq 0.90$ (with some as high as $q = 0.97$) and a minimum element quality measure of $q > 0.30$. Note that a $q = 0.30$ corresponds to an element with a minimum angle of 20°. Results for a set of example meshes created using ADMESH are presented in the following subsections. Mesh statistics are summarized in Table 1.

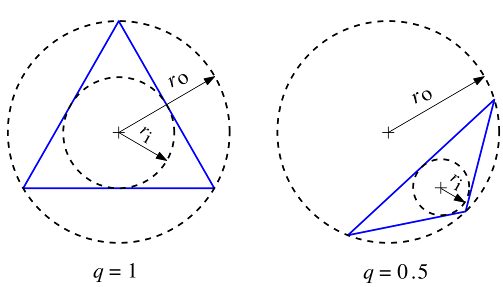


**Fig. 9** Geometric depiction of the element quality measure used to assess mesh quality

### 3.2 The Western North Atlantic Tidal domain

The computational domain shown in Figs. 8 and 10—often referred to as the Western North Atlantic tidal (WNAT) domain (Hagen and Parrish 2004)—has been used in a number of tidal and hurricane storm surge propagation studies; see, for example, Dawson et al. (2011), Hagen et al. (2002), Hagen and Parrish (2004), Leuttich and Westerink (1995), Westerink et al. (1994a, b). In addition to encompassing much of the Western North Atlantic, the WNAT domain also includes the Gulf of Mexico and the Caribbean Sea. It has an open-ocean boundary along the 60° W meridian and coastal boundaries primarily consist of South, Central, and North American coastlines. Given the large extent of the WNAT domain and the need to provide high resolution in coastal regions in order to adequately resolve geometry and tidal flow, the use of an unstructured mesh is particularly advantageous. High resolution can be provided in areas of relatively shallow water and rapidly changing bathymetric gradients, while lower, though still adequate, resolution can be used in the deeper ocean.

Starting with boundary and bathymetric data obtained from an existing mesh of the WNAT domain, an unstructured mesh was generated automatically using ADMESH and it is shown in Figs. 8 and 10. The resulting mesh consists of 96,751 elements and 49,918 nodes, with a maximum element size of 200 km in the open ocean and a minimum element size of 5 km in the near-shore and shelf-break regions. The mesh was generated in 4.3 min and has a mean mesh quality measure of $q = 0.97$, and a minimum mesh quality measure of $q = 0.32$. It can be noted how high resolution is automatically provided along the shelf break of the Western North Atlantic, the intricate coastlines of the Americas, and around Bermuda and the numerous islands in the Caribbean Sea. While this mesh does not possess the resolution in the near-shore regions to perform accurate calculations such as hurricane storm surge predictions, it is sufficient for tidal simulations, as it is comparable in terms of the resolution it provides to

**Table 1** Summary of the meshes generated with ADMESH

| Domain | No. of nodes | Run time (h) | Max. element size (km) | Min. element size (m) | Mean and min. $q$ | |
|---|---|---|---|---|---|---|
| WNAT | 49,918 | 0.16 | 200 | 5,000 | 0.9734 | 0.3170 |
| Gulf of Mexico | 144,240 | 1.46 | 50 | 200 | 0.9662 | 0.3215 |
| Cape Cod | 167,790 | 0.56 | 2 | 25 | 0.9723 | 0.3173 |
| UK & Ireland | 317,375 | 0.71 | 50 | 500 | 0.9064 | 0.3836 |
| Italy | 13,673 | 0.01 | 100 | 2,500 | 0.9306 | 0.3451 |

a 53,000-node mesh used for tidal validation reported in Hagen et al. (2004), which was generated using the LTEA approach. This type of mesh could also be appended with a near-shore high resolution mesh, such as the Gulf of Mexico and Cape Cod meshes discussed below, to perform more detailed simulations in local coastal regions.

### 3.3 The gulf of Mexico

A more detailed mesh of a large portion of the Gulf of Mexico was also created using ADMESH and can be seen in Fig. 11, where the insets of the figure show how well ADMESH captures the intricacy of the coastline and the barrier islands, along with major changes in the bathymetry in the near-shore and shelf-break regions. This mesh consists of 273,432 elements and 144,240 nodes—the majority of which are clustered in the near-shore and shelf-break regions. The mesh has a maximum element size of 50 km in the open waters of the Gulf and a minimum element size of 200 m along the shoreline. The mesh was generated in 1.46 h and is an example of a high-resolution mesh that could be appended to the WNAT mesh shown in Fig. 8 to perform high resolution simulations. We note that the run time for this particular mesh is substantially higher than the WNAT mesh due to the fact that run times are directly proportional to the ratio of the largest

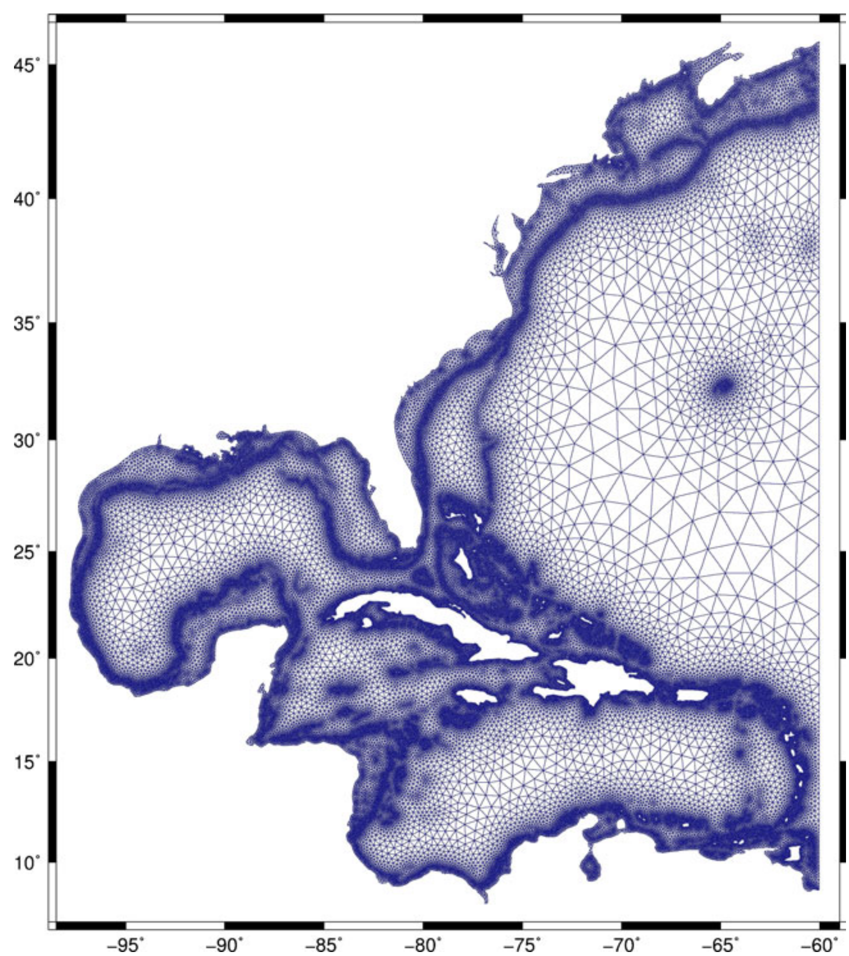


**Fig. 10** A mesh of the Western North Atlantic Tidal domain automatically generated by ADMESH

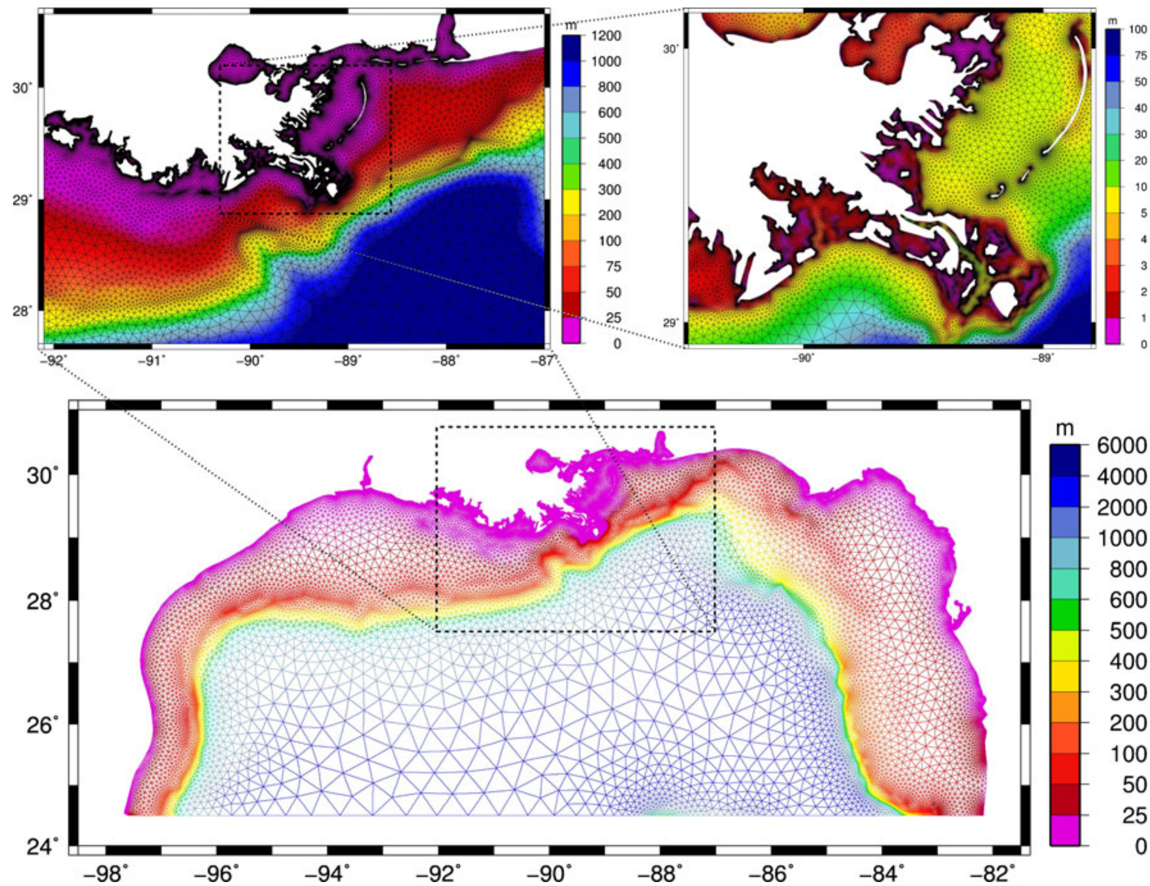


**Fig. 11** A mesh of the Gulf of Mexico automatically generated by ADMESH. *Color bars* indicate bathymetric depth in meters

horizontal dimension of the domain and the minimum element size specified (the ratio for this case is approximately 10,000:1 compared to approximately 1,000:1 for the WNAT mesh). The larger this ratio, the larger the background grid, and thus the more calculations that must be performed to initialize the distance function, extract local feature sizes, and solve the gradient limiting Eq. 21.

## 3.4 UK and Ireland

A domain around the UK and Ireland was also generated using ADMESH, as shown in Fig. 12. The rapidly changing bathymetry in this region, along with the intricate shoreline of the UK and Ireland makes for an interesting test case for ADMESH. As can be seen in Fig. 12, the mesh created uses high resolution in the near shore and in areas of steep bathymetric change, such as the shelf break where the element size is 1 km, and larger elements in areas that are less likely to develop interesting flow characteristics, such as in the area to the west of the shelf break where element sizes

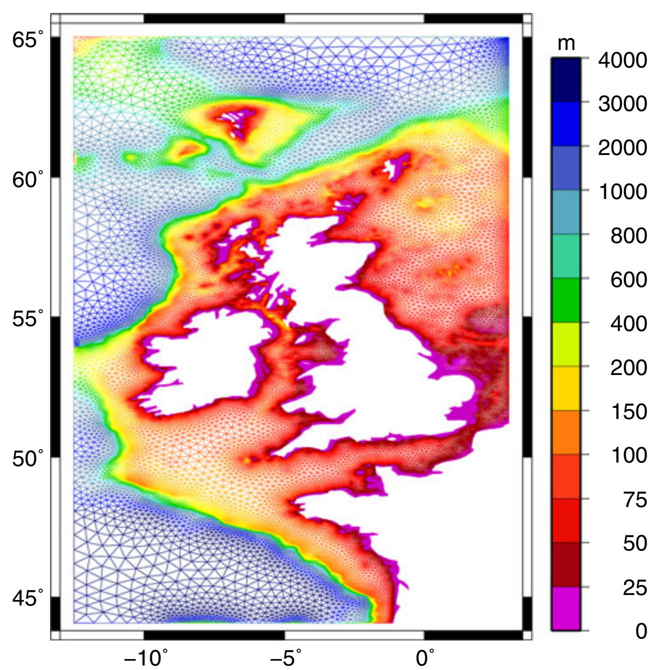


**Fig. 12** A mesh around the UK and Ireland automatically generated by ADMESH. *Color bars* indicate bathymetric depth in meters

reach 50 km. The mesh consists of 618,807 elements and 317,375 nodes, was created in 42.86 min, and has a mean mesh quality measure of $q = 0.91$ and a minimum mesh quality measure of $q = 0.38$.

### 3.5 Meshes without bathymetry

As noted earlier, ADMESH can be run with any of the mesh size arguments turned off. For the last two examples—a refined mesh around Cape Cod, Massachusetts and a mesh around Italy, which are shown in Figs. 13 and 14, respectively—we turn off the bathymetric mesh size requirements in order to highlight the ability of ADMESH to resolve intricate coastlines. This can perhaps especially be seen in the mesh of Cape Cod where the inset of Fig. 13 shows how well the intricacies of Little Pleasant Bay are automatically resolved by ADMESH. These two examples also highlight the ability of ADMESH to generate meshes that are properly graded in moving from geometrically complex coastlines out into the open ocean.

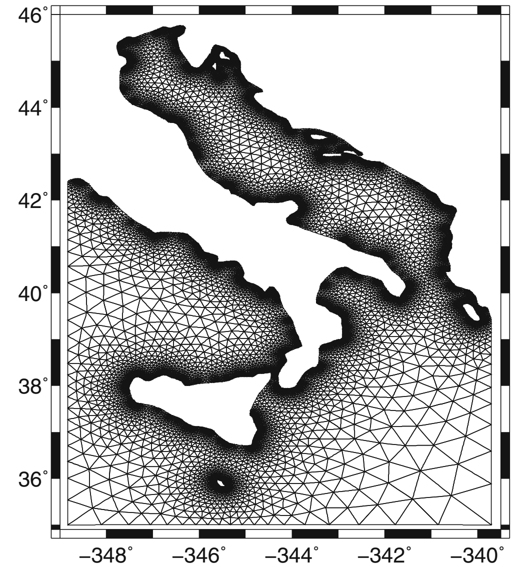


**Fig. 14** A mesh of Italy automatically generated by ADMESH with $g = 0.20$

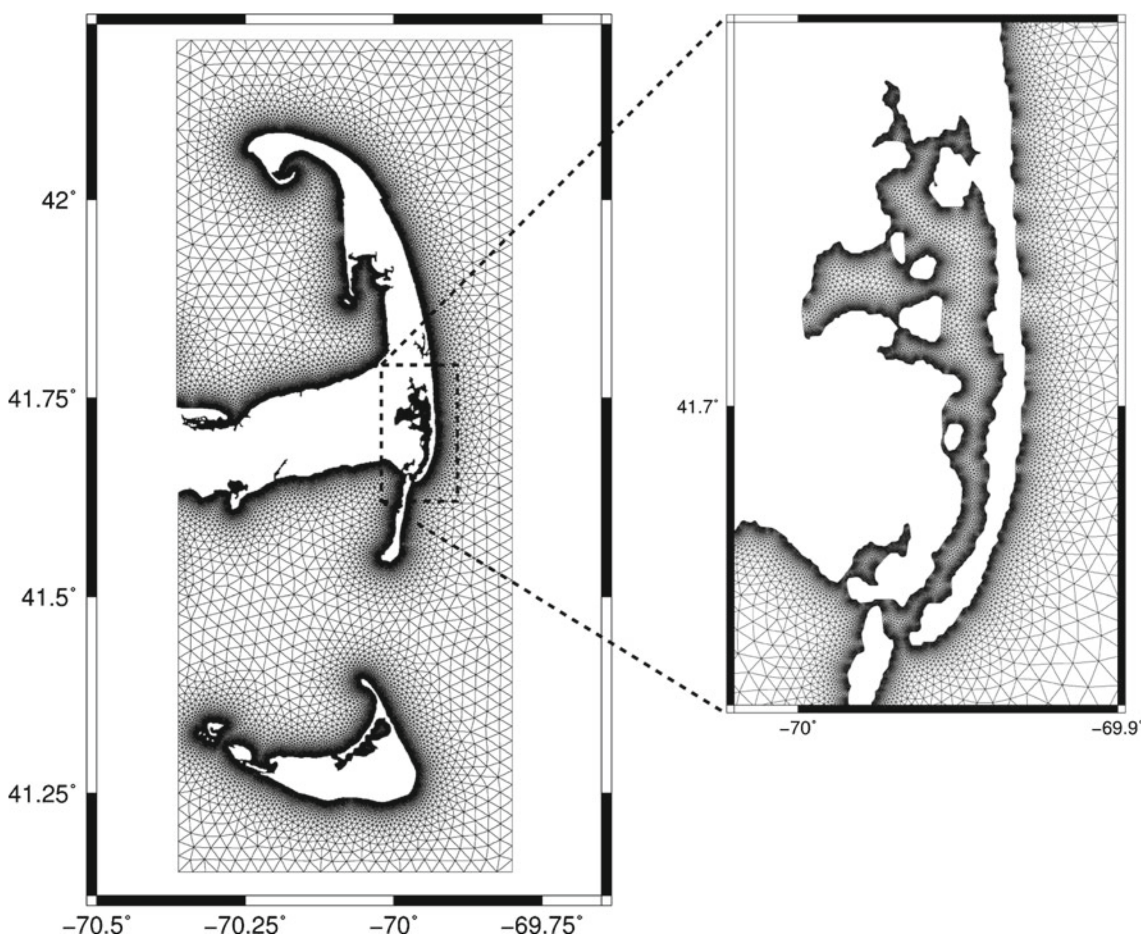


**Fig. 13** A mesh of Cape Cod automatically generated by ADMESH, with the *inset* showing a close-up of Little Pleasant Bay

## 4 Summary and future work

ADMESH automatically generates unstructured triangular meshes for shallow water domains using minimal user input. The use of a mesh size function $h(\mathbf{x})$ that prescribes local element sizes based on a number of physical and geometric factors along with the grading adaptation and a spring-based force equilibrium approach ensures that the meshes produced by ADMESH are of high quality and possess the appropriate amount of resolution necessary throughout the domain. The run times for ADMESH are generally less than an hour but are highly dependent on the size of the domain and the minimum element size specified by the user. Specifically, run times are directly proportional to the ratio of the largest horizontal dimension of the domain and the minimum element size specified. Additionally, the number of nodes in meshes created by ADMESH is dependent on the parameter values input by the user; for example, larger values of $g$ result in a mesh with fewer elements (due to coarser mesh grading), while higher $K$ values result in a mesh with more elements. With regard to future developments, the structure of the ADMESH program is such that it allows for additional arguments to be added to the mesh size function; for example, alternative bathymetry/tidal constraints, a posteriori error estimates, such as those from LTEA, etc., which make ADMESH a highly flexible program. Two specific fronts along which future developments will focus are as follows:

1. *Accurate coastline representation*: Coastline data is currently obtained by ADMESH either from an existing mesh of the area of interest or from NOAA's online Coastline Extractor tool. It would be beneficial to allow users to directly extract coastlines from available satellite images or from Google Earth. Therefore, future development will focus on coupling ADMESH with available image segmentation software. In addition to the use of image segmentation software, work has already begun that will allow ADMESH to construct coastline data from a given set of bathymetry data, thus, allowing meshes to be created solely from the bathymetry of the shallow water domain.
2. *Mixed element meshes*: ADMESH currently produces meshes consisting solely of triangular elements. It has been shown by Maggi (2011) that meshes consisting of both triangular and quadrilateral elements, i.e., mixed element meshes, are more efficient while offering comparable accuracy to meshes composed solely of triangular elements for certain classes of problems. Therefore, it would be advantageous for ADMESH to create meshes that consist of quadrilaterals in the deep ocean, where the geometry and solutions are relatively homogeneous, and triangles in the near-shore regions, where complex shorelines and flow fields dictate spatially varying resolution.

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